

Marius Schneider

Postdoctoral Researcher
Computational neuroscience | Active vision
[Website](#) | [GoogleScholar](#) | [Github](#) | [ORCID](#)

Education

Ph.D. in Neurophysics <i>Donders Centre for Neuroscience, Radboud University</i>	02/2020–05/2024 Nijmegen, Netherlands
Thesis: <i>Mechanisms of inter-areal neuronal communication</i> . Advisor: Prof. Dr. Martin Vinck. Committee: Prof. Dr. Marie Carlen, Prof. Dr. Gaute Einevoll, Prof. Dr. Dr. h.c. mult. Wolf Singer. Graduated with highest honors (top 5%).	
M.Sc. in Physics <i>Goethe University Frankfurt</i>	10/2016–04/2019 Germany
Thesis: <i>Biological complexity facilitates tuning of the neuronal parameter space</i> . Advisors: Dr. Hermann Cuntz, Prof. Dr. Peter Jedlicka. Grade: 1.1 (GPA: 3.9).	
B.Sc. in Physics <i>Goethe University Frankfurt</i>	10/2012–10/2016 Germany

Professional Experience

Postdoctoral Researcher <i>UC Santa Barbara</i>	2024–Present California, USA
Developing digital-twin models of active vision and Mouse vs AI closed-loop benchmarks for robust embodied vision.	
PhD Student <i>Ernst Strüngmann Institute for Neuroscience - Max Planck Society</i>	2019–2024 Frankfurt, Germany
Research Assistant <i>Justus Liebig University</i>	2018–2019 Gießen, Germany
Research Assistant <i>Frankfurt Institute for Advanced Studies</i>	2017–2018 Frankfurt, Germany
Accelerator Operator <i>Goethe University</i>	2017–2018 Frankfurt, Germany
Research Assistant <i>Max Planck Institute for Empirical Aesthetics</i>	2016–2018 Frankfurt, Germany

Awards, Fellowships, and Grants

• COSYNE Travel Grant	2026
• EBBS Young Investigator Award, European Brain and Behaviour Society	2024
• PhD Research Fellowship, International Max Planck Research School for Neural Circuits	2019
• Travel Grant for CNS Conference, Organization for Computational Neurosciences	2019
• Travel Grant for Neural Dynamics Summer School, University of Bristol	2018
• German National Student Scholarship	2016

Publications

* denotes equal contribution

- Onorato, I., Tzanou, A., **Schneider, M.**, Uran, C., Broggin, A. C., & Vinck, M. (2025). Distinct roles of PV and Sst interneurons in visually-induced gamma oscillations. *Cell Reports*, 44(3), 115385.
- Spyropoulos, G.*, **Schneider, M.***, van Kempen, J., Gieselmann, M. A., Thiele, A., & Vinck, M. (2024). Distinct feedforward and feedback pathways for cell-type specific attention effects.

Neuron, 112(14), 2423–2434.e7.

3. **Schneider, M.**, Tzanou, A., Uran, C., & Vinck, M. (2023). Cell-type-specific propagation of visual flicker. *Cell Reports*, 42(5).
4. **Schneider, M.**, Bird, A. D., Gidon, A., Triesch, J., Jedlicka, P., & Cuntz, H. (2023). Biological complexity facilitates tuning of the neuronal parameter space. *PLOS Computational Biology*, 19(7), e1011212.
5. Vinck, M., Uran, C., Spyropoulos, G., Onorato, I., Broggin, A. C., **Schneider, M.**, & Johnson, A. C. (2023). Principles of large-scale neural interactions. *Neuron*, 111(7), 987–1002.
6. Dowdall, J. R., **Schneider, M.**, & Vinck, M. (2023). Attentional modulation of inter-areal coherence explained by frequency shifts. *NeuroImage*, 277, 120256.
7. **Schneider, M.**, Broggin, A. C., Dann, B., Tzanou, A., Uran, C., Sheshadri, S., Scherberger, H., & Vinck, M. (2021). A mechanism for inter-areal coherence through communication based on connectivity and oscillatory power. *Neuron*, 109(24), 4050–4067.
8. Cuntz, H., Bird, A., Beining, M., **Schneider, M.**, Mediavilla, L., Hoffmann, F., Deller, T., & Jedlicka, P. (2021). A general principle of dendritic constancy – a neuron’s size and shape invariant excitability. *Neuron*, 109(22), 3647–3662.

Under review

1. Lima, A.* , Hou, Y.*, Beyeler, M., **Schneider, M.** (2025). Beyond Neural Activity Prediction: Probing Latent Representations in Mouse V1 Digital Twins. *In review at NeurIPS*.
2. **Schneider, M.**, Canzano, J., Peng, J., Hou, Y., Smith, S. L., & Beyeler, M. (2025). Mouse vs. AI: A neuroethological benchmark for visual robustness and neural alignment. *In review at NeurIPS*.
3. Vinck, M., Uran, C., & **Schneider, M.** Aperiodic processes explaining rhythms in behavior: A matter of false detection or definition? *In review*.

Invited Talks and Selected Conference Presentations

- Poster: Predictive models trained on natural behavior recover cell- and state-dependent tuning in mouse V1. COSYNE 2026. Lisbon, Portugal. 03/2026.
- Poster: Uncovering behavioral strategies: Training mice and AI on a shared foraging task. COSYNE 2025. Montreal, Canada. 03/2025.
- Poster: A mechanism for selective attention in biophysically realistic Daleian spiking neural networks. COSYNE 2025. Montreal, Canada. 03/2025.
- Poster: A deep learning framework for center-periphery visual processing in mouse visual cortex. COSYNE 2025. Montreal, Canada. 03/2025.
- Poster: Mechanisms of attention in biophysiologically realistic Daleian spiking neural networks. FENS Forum. Vienna, Austria. 2024.
- Invited Talk: Do neurons communicate through coherence? Bernstein Center of Computational Neurosciences. Berlin, Germany. 2022.
- Poster: Cell-type specific entrainment during rhythmic visual flicker stimulation. SFN. San Diego, USA. 2022.
- Selected Talk: A mechanism for inter-areal coherence through communication based on connectivity and oscillatory power. Neuromatch Conference. Online. 2021.

Teaching and Mentoring

- Guest Lecturer, Graduate Course: Bionic Vision UC Santa Barbara, 2026
- Guest Lecturer, Graduate Course: Bionic Vision UC Santa Barbara, 2025

- Master's Thesis Supervision: Jing Peng UC Santa Barbara, 2024
- TA, Neuromatch Academy: Computational Neuroscience Online, 2022
- Bachelor Thesis Supervision: Aysin Yildirim Goethe University, 2019
- Teaching Assistant, 7th Baltic-Nordic School on Neuroinformatics FIAS, 2019
- Teaching Assistant, Computational Neurobiology Course Goethe University, 2018

Academic Contributions and Services

- **Conference Organization:** Lead organizer of NeurIPS 2025 Competition & Workshop: *Mouse vs AI – Robust Visual Foraging Challenge*.
- **Seminar Organization:** Co-organizer of lecture series ESI-talks, Ernst Strüngmann Institute for Neuroscience.
- **Peer Review:** NeurIPS, Nature Communications, PLOS Computational Biology, Journal of Neurophysiology, Frontiers in Computational Neuroscience.